

PURA Syndrome: Comprehensive Disease Characteristics Report

Disease: PURA Syndrome (PURA-related neurodevelopmental disorder) **Identifiers:** MONDO:0014601 · OMIM #616158 · Orphanet ORPHA:438213 · Gene: *PURA* (HGNC:9701), 5q31.3
Category: Mendelian, autosomal dominant, de novo

Summary

PURA syndrome is a rare, highly penetrant, autosomal dominant neurodevelopmental disorder caused by **de novo heterozygous loss-of-function or protein-misfolding variants in *PURA***, the gene at chromosome 5q31.3 encoding the DNA/RNA-binding protein **Pur-alpha (Purα)**. The disorder was defined in 2014 when two independent whole-exome trio studies established that de novo *PURA* variants recapitulate the severe neurological phenotype previously attributed to the larger 5q31.3 microdeletion syndrome, identifying *PURA* as the phenocritical gene ([PMID: 25439098](#); [PMID: 25342064](#)). Pur-alpha is essential for postnatal neuronal proliferation, developmentally-timed dendrite formation, and the dendritic transport and local translation of mRNA; haploinsufficiency of this function disrupts brain development and produces the clinical syndrome.

The core clinical picture is one of a **multisystem disorder dominated by neurological features**: profound neonatal hypotonia, early feeding difficulties, global developmental delay progressing to moderate-to-severe intellectual disability with **near-universal absence of speech**, frequent non-ambulation, seizures and abnormal non-epileptic movements, hypersomnolence, and neonatal problems including excessive hiccups, recurrent apnea, and hypothermia. Less common but well-documented manifestations include congenital heart defects, urogenital malformations, and skeletal, ophthalmological, gastrointestinal, and endocrine anomalies ([PMID: 33275834](#)). As of 2024, approximately **650 patients worldwide** had been reported ([PMID: 38655849](#)).

Diagnosis is molecular — via whole-exome/genome sequencing, targeted *PURA* sequencing (the syndrome is clinically recognizable), or chromosomal microarray for 5q31.3 deletions. There is **no reliable genotype-phenotype correlation** and **no disease-modifying therapy**;

management is lifelong, supportive, and multidisciplinary. Recurrence risk to siblings is low (~1%), attributable to possible parental germline mosaicism. The main life-limiting risk is respiratory complication, which caused death in the oldest reported patient at age 26 ([PMID: 26582469](#)).

1. Disease Information

Overview. PURA syndrome is a Mendelian neurodevelopmental disorder characterized by moderate-to-severe developmental delay/intellectual disability, neonatal hypotonia, feeding difficulties, hypersomnolence, seizures or seizure-like movements, and a range of multisystem anomalies. It was first delineated as a distinct entity in 2014 and represents the phenocritical core of the previously described 5q31.3 microdeletion syndrome.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0014601 (PURA-related severe neurodevelopmental disorder). Note: the template-supplied MONDO:1060108 is not the current primary ID.
OMIM	#616158 (Neurodevelopmental disorder with neonatal respiratory insufficiency, hypotonia, and feeding difficulties, NEDRIHF)
Orphanet	ORPHA:438213
Gene	<i>PURA</i> , HGNC:9701, NCBI Gene 5813, UniProt Q00577
Locus	5q31.3
ICD-10 / ICD-11	No specific code; maps under developmental anomaly / G-code categories

Synonyms / alternative names. PURA-related neurodevelopmental disorder; PURA-related severe neurodevelopmental disorder with neonatal respiratory insufficiency, hypotonia, and feeding difficulties (NEDRIHF); Pur-alpha syndrome; the phenocritical core of "5q31.3 microdeletion syndrome."

Information source. The disease-level knowledge is derived from **aggregated case series and cohort studies** (parent-child trio exome studies, multi-center clinical cohorts of 25–32 individuals, and international reviews), not from individual EHR mining. Patient registries maintained by the PURA Syndrome Foundation support ongoing aggregation.

2. Etiology

Primary cause — genetic. PURA syndrome is caused by **de novo heterozygous loss-of-function variants in PURA**, or by deletions of 5q31.3 that encompass *PURA*. Two independent 2014 whole-exome trio studies established causation:

- Lalani et al. described **11 individuals** with clinical features of 5q31.3 microdeletion syndrome carrying de novo *PURA* mutations within the critical region, concluding: *"We describe 11 individuals with clinical features of 5q31.3 microdeletion syndrome and de novo mutations in PURA, encoding transcriptional activator protein Pur-α, within the critical region. These data implicate causative PURA mutations responsible for the severe neurological phenotypes observed in this syndrome"* (PMID: 25439098).
- Hunt et al. (Deciphering Developmental Disorders study) identified **protein-altering de novo mutations in four trios**: *"Protein-altering de novo mutations in PURA were identified in four subjects. They include two different frameshifts, one inframe deletion and one missense mutation"* (PMID: 25342064).

That *PURA* point mutations recapitulate the severe neurological phenotype of the larger 5q31.3 microdeletion confirms *PURA* as the phenocritical gene: *"The recent finding that de novo PURA point mutations are indeed sufficient to cause the severe neurological symptoms also observed in patients with 5q31.2q31.3 deletion further reinforces the gene's causative role in 5q31.3 microdeletion syndrome"* (PMID: 26582469).

Genetic risk factors. The causal variant is essentially the sole determinant. Variants are almost always de novo (arising in the affected individual, not inherited). No susceptibility loci or established modifier genes have been identified. Disruption of Pur repeat III has been proposed to worsen outcome, but no robust modifier has been validated (PMID: 25342064).

Environmental risk factors. None identified. Because variants are predominantly de novo, advanced parental age is a general (nonspecific) contributor to de novo mutation rate, but no specific environmental exposure, toxin, or lifestyle factor has been linked to PURA syndrome.

Protective factors. None identified (genetic or environmental).

Gene-environment interactions. None documented. PURA syndrome is a monogenic disorder with high penetrance; the phenotype is driven by the genetic lesion rather than gene-environment interaction.

3. Phenotypes

PURA syndrome is a multisystem disorder with a strong neurological predominance. Phenotype frequencies are drawn primarily from the 25-patient Chinese cohort ([PMID: 36376392](#)), the 32-individual delineation ([PMID: 29097605](#)), and the international review ([PMID: 33275834](#)).

Phenotype	Type	Onset	Frequency	Suggested HPO term
Global developmental delay / intellectual disability	Neurodevelopmental	Infancy	~Universal (moderate–severe)	HP:0001263 / HP:0001249
Neonatal hypotonia	Clinical sign	Neonatal	Very frequent (near-universal)	HP:0001319 / HP:0001252
Absent/severely limited speech	Neurodevelopmental	Childhood	Nearly all patients	HP:0001344 (absent speech)
Feeding difficulties	Clinical sign	Neonatal	Very frequent	HP:0011968
Seizures / seizure-like movements	Neurological	Neonatal–childhood	Common	HP:0001250
Abnormal non-epileptic movements	Neurological	Early	Common	HP:0002072 (dyskinesia)
Hypersomnolence / excessive sleepiness	Behavioral/neuro	Neonatal–infancy	Prominent	HP:0002360 / HP:0100786
Excessive hiccups	Clinical sign	Neonatal	Characteristic early sign	HP:0100585 (hiccough)
Recurrent apnea	Clinical sign	Neonatal	Frequent	HP:0002104 (apnea)
Hypothermia / temperature dysregulation	Clinical sign	Neonatal	Frequent	HP:0002045
Non-ambulation / motor impairment	Physical	Childhood	Many	HP:0002505 (inability to walk)
Dysmorphic facial features	Physical	Congenital	Frequent (more in deletion carriers)	HP:0001999

Phenotype	Type	Onset	Frequency	Suggested HPO term
Congenital heart defects	Physical malformation	Congenital	Less common	HP:0001627
Urogenital malformations	Physical malformation	Congenital	Less common	HP:0000119
Ophthalmological anomalies	Physical	Variable	Less common	HP:0000478
Skeletal anomalies	Physical	Variable	Less common	HP:0000924
Endocrine anomalies (e.g., pituitary dysregulation, pubertal delay)	Laboratory/clinical	Variable	Less common	HP:0000818

Characteristics. Onset is overwhelmingly **neonatal to early infantile**. Severity is **moderate to severe** and variable. The course is chronic and largely **stable/non-progressive** in most, though at least one patient with a *PURA*-encompassing deletion showed adolescent-onset neurological deterioration, pubertal delay/primary amenorrhea, and progressive respiratory decline leading to death at 26 ([PMID: 26582469](#)).

Key supporting quotes: - *"Developmental delay/intellectual disability, neonatal hypotonia, neonatal feeding difficulties, hypersomnolence and dysmorphic features were prominent clinical features in PURA syndrome"* ([PMID: 36376392](#)). - *"moderate to severe neurodevelopmental delay with absence of speech in nearly all patients and lack of independent ambulation in many. Early-onset problems include excessive hiccups, hypotonia, hypersomnolence, hypothermia, feeding difficulties, recurrent apneas, epileptic seizures, and abnormal nonepileptic movements. Other less common manifestations comprise congenital heart defects, urogenital malformations, and various skeletal, ophthalmological, gastrointestinal, and endocrine anomalies"* ([PMID: 33275834](#)). - *"Neonatal hypotonia, early feeding difficulties and seizures, or 'seizure-like' movements, were also common"* ([PMID: 25342064](#)).

Quality of life impact. Profound. Most patients are non-verbal and many are non-ambulatory, requiring lifelong caregiving, augmentative communication, feeding support, and management of seizures and sleep. Daily functioning is severely affected across communication, mobility, and self-care domains.

4. Genetic / Molecular Information

Causal gene. *PURA* (HGNC:9701; UniProt Q00577), a single-exon gene at 5q31.3 encoding Pur-alpha, a purine-rich element binding protein composed of repeated PUR domains.

Pathogenic variants. - **Variant types:** frameshift (e.g., c.159dup p.(Leu54Alafs147)), *in-frame deletions* (recurrent c.697_699del p.(Phe233del)), nonsense, missense, and whole-gene deletions (5q31.3 microdeletion). Both initial trios and later cohorts show a spectrum consistent with loss of function ([PMID: 25342064](#); [PMID: 36376392](#)). - **Classification (ACMG/AMP):** the vast majority are pathogenic/likely pathogenic. As of 2021, 64 different pathogenic variants across 78 individuals were reported ([PMID: 33275834](#)). - **Recurrent variants:** c.697_699del p.(Phe233del) and c.159dup p.(Leu54Alafs147) ([PMID: 36376392](#); [PMID: 37204304](#)). - **Allele frequency:** essentially absent from population databases (gnomAD) — consistent with de novo, highly penetrant, severe disease. - **Origin:** germline, de novo. Somatic/mosaic events are rare but relevant to recurrence risk. - **Functional consequence: loss of function / haploinsufficiency.** *PURA*-syndrome mutations impair the protein's 3D folding, disrupt RNA binding, and cause failure of *PURA* to localize to P-bodies ([PMID: 38655849](#)).

Modifier genes. None validated. Disruption of Pur repeat III has been proposed to worsen outcome ([PMID: 25342064](#)), but no reliable genotype-phenotype correlation exists ([PMID: 37204304](#); [PMID: 29097605](#)).

Epigenetic information. No disease-specific DNA-methylation episignature has been established for *PURA* syndrome to date (not available).

Chromosomal abnormalities. The 5q31.3 (5q31.2q31.3) microdeletion encompassing *PURA* produces a phenotype overlapping point-mutation cases; the smallest reported de novo deletion involving *PURA* helped confirm the gene's centrality ([PMID: 26582469](#); [PMID: 25439098](#)).

5. Environmental Information

- **Environmental factors:** None known to cause or modify *PURA* syndrome. It is a monogenic de novo disorder.
- **Lifestyle factors:** Not applicable.

- **Infectious agents:** Not applicable. (Of note, host Pur-alpha is exploited by certain viruses — HIV-1, JC virus — and a baculovirus encodes a PUR α -like protein VP1054 [[PMID: 23720732](#)] — but these are not causes of PURA syndrome.)

6. Mechanism / Pathophysiology

Ordered causal chain

1. A **de novo heterozygous loss-of-function variant** (frameshift, nonsense, in-frame deletion, missense) or a **5q31.3 deletion** occurs in *PURA* → **leads to** reduced dosage and/or misfolded Pur-alpha protein (haploinsufficiency).
2. Mutations **impair the 3D folding of the PUR domains** → **result in** loss of the PC4-like β - β - α fold that mediates single-stranded DNA/RNA binding and duplex unwinding ([PMID: 38655849](#); [PMID: 26744780](#); [PMID: 32476604](#)).
3. Misfolding **disrupts Pur-alpha's nucleic-acid binding** → **causes** failure of Pur-alpha to associate with P-bodies and RNP granules ([PMID: 38655849](#)).
4. Loss of RNP/P-body function → **impairs** dendritic mRNA transport and local translation (Pur-alpha normally operates with **Staufen, FMRP, and BC1/BC200 non-coding RNA** at dendritic translation sites) ([PMID: 16511857](#)).
5. Disrupted mRNA localization and translation → **impairs** developmentally-timed **dendrite formation** and **postnatal neuronal proliferation** (inferred from mouse loss-of-function models) ([PMID: 16511857](#); [PMID: 22010047](#)).
6. Abnormal neuronal proliferation and dendritic maturation → **result in** reduced neuron numbers and dendritic deficits (demonstrated in *Pura* +/- mice: reduced neurons in cerebellar vermis and hippocampal CA1–3) ([PMID: 27651147](#)).
7. Structural/functional CNS deficits → **manifest as** neonatal hypotonia, feeding difficulties, developmental delay/intellectual disability, absent speech, seizures, movement abnormalities, hypersomnolence, and brainstem-related dysregulation (apnea, hypothermia, hiccups) ([PMID: 33275834](#); [PMID: 25342064](#)).

Branch: Pur-alpha also binds guanine-rich/G-quadruplex nucleic-acid elements; loss of this activity is a candidate contributor to translational dysregulation (mechanistically related to Pur biology in C9orf72 ALS/FTD, where increased *Pura* is protective) ([PMID: 32035967](#)). Its relevance to PURA syndrome specifically is **inferred, not demonstrated**.

Detail by category

Molecular pathways / protein function. Pur-alpha is a repeated **PUR-domain** protein; each ~60–80 aa domain adopts a PC4-like fold that binds ssDNA/RNA and can unwind duplex nucleic acids. The crystal structure of the DNA/RNA-binding domain in complex with ssDNA *"reveals base-specific recognition and offers a molecular explanation for the effect of point mutations in the 5q31.3 microdeletion syndrome"* and identified a highly conserved phenylalanine essential for unwindase activity ([PMID: 26744780](#)). The domain family is described in [PMID: 32476604](#).

Cellular processes. Neuronal precursor proliferation, dendrite formation, and mRNA transport/local translation. *"the protein Pura ... is essential for neuronal proliferation, dendritic maturation, and the transportation of mRNA to translation sites"* ([PMID: 37204304](#)).

Protein dysfunction. *"the mutations impair the protein's folding and thereby disrupt PURA's ability to bind RNA"* and *"in cells carrying PURA syndrome mutations, PURA failed to move adequately to P-bodies"* ([PMID: 38655849](#)). This is a loss-of-function mechanism via misfolding.

Dendritic mRNA transport machinery. *"Pur alpha and Staufen are colocalized at dendritic sites of mRNA translation,"* and *"double-RNA immunoprecipitation places Pur alpha together with Staufen or FMRP on BC1 RNA and specific mRNA species in vivo"* ([PMID: 16511857](#)). Microtubule disruption mislocalizes Pur-alpha to axons, indicating microtubule-dependent dendritic targeting.

Suggested GO / CL terms. - Biological processes: GO:0016358 (dendrite development), GO:0050767 (regulation of neurogenesis), GO:0006417 (regulation of translation), GO:0008298 (intracellular mRNA localization), GO:0006355 (regulation of DNA-templated transcription). - Molecular function: GO:0003697 (single-stranded DNA binding), GO:0003727 (single-stranded RNA binding). - Cellular components: GO:0000932 (P-body), GO:0030425 (dendrite), GO:0005634 (nucleus). - Cell types: CL:0000540 (neuron), CL:0000121 (cerebellar/Purkinje neuron), CL:0002608 (hippocampal neuron), CL:0011020 (neural progenitor cell).

7. Anatomical Structures Affected

Organ level. - **Primary:** Brain / central nervous system (UBERON:0000955 brain; UBERON:0002037 cerebellum; UBERON:0002421 hippocampal formation; brainstem UBERON:0002298). - **Secondary / less common:** heart (congenital defects, UBERON:0000948), urogenital system (UBERON:0000990), skeleton (UBERON:0004288), eye (UBERON:0000970), gastrointestinal tract (UBERON:0001555), endocrine system including pituitary

(UBERON:0000007). - **Body systems:** nervous (predominant), musculoskeletal (hypotonia), respiratory (apnea, later restrictive decline), autonomic/thermoregulatory (hypothermia), endocrine.

Tissue and cell level. Nervous tissue is centrally affected. Cerebellar vermis neurons and hippocampal pyramidal neurons (CA1–CA3) show reduced numbers in the heterozygous mouse model ([PMID: 27651147](#)). Cell Ontology: CL:0000540 (neuron), CL:0000121 (Purkinje/cerebellar neuron), CL:0002608 (hippocampal neuron), CL:0011020 (neural progenitor cell).

Subcellular level. Dendrites (GO:0030425), cytoplasmic P-bodies / mRNP granules (GO:0000932), and the nucleus (GO:0005634). Pur-alpha is specifically located in dendrites colocalized with MAP2 — not in axons ([PMID: 16511857](#)).

Localization / lateralization. CNS involvement is bilateral and generalized. Brain MRI abnormalities are more common in 5q31.3-deletion carriers than in point-mutation patients ([PMID: 36376392](#)); findings can include nonspecific white-matter and delayed-myelination changes.

8. Temporal Development

- **Onset:** Congenital / neonatal. Hypotonia, feeding difficulty, hiccups, apnea, hypothermia, and hypersomnolence present in the newborn period; developmental delay becomes apparent in infancy.
- **Onset pattern:** Chronic/insidious in most; the disorder is present from birth.
- **Progression:** Largely **stable/non-progressive** neurodevelopmental disability across most patients — delay is fixed rather than regressive. However, long-term follow-up documented **adolescent-onset deterioration** with pubertal delay/primary amenorrhea and progressive respiratory decline in a deletion patient, indicating that a progressive course can occur in some ([PMID: 26582469](#)).
- **Disease course pattern:** Chronic, lifelong. Seizures may be episodic. Neonatal features (hiccups, apnea, hypothermia, hypersomnolence) tend to attenuate over the first months to years while cognitive/motor disability persists.
- **Duration:** Lifelong.
- **Critical periods:** The postnatal window of neuronal proliferation and dendritic maturation is the biologically critical period; this is when Pur-alpha function is most essential (inferred from mouse developmental phenotypes, [PMID: 22010047](#); [PMID: 16511857](#)).

9. Inheritance and Population

Epidemiology. No formal prevalence/incidence has been established. Reported patient numbers grew from 4 (2014) → 32 (PMID: 29097605) → 78 with 64 variants (2021) (PMID: 33275834) → ~650 worldwide (2024) (PMID: 38655849). *"PURA syndrome is a neurodevelopmental disorder that affects about 650 patients worldwide"* (PMID: 38655849). PURA syndrome is a recognized contributor to unexplained neonatal hypotonia and early-onset developmental delay.

Genetic etiology parameters. - **Inheritance:** Autosomal dominant, almost always de novo. - **Penetrance:** High / near-complete — nearly every reported mutation causes symptoms (PMID: 38655849). - **Expressivity:** Variable, wide-ranging clinical spectrum (PMID: 37204304). - **Genetic anticipation:** Not applicable (not a repeat-expansion disorder). - **Germline mosaicism:** Possible; underlies the low (~1%) sibling recurrence risk. - **Founder effects / consanguinity:** Not applicable (de novo dominant). - **Carrier frequency:** Not applicable.

Population demographics. No ethnic predilection; cases reported worldwide (Europe, North America, China, and elsewhere). No strong sex bias is established. Age distribution reflects pediatric ascertainment, though affected individuals survive into adulthood.

10. Diagnostics

Recommended approach. Molecular genetic testing is the diagnostic cornerstone. The original disease-defining diagnoses were made by **whole-exome sequencing of parent-child trios** (PMID: 25342064; PMID: 25439098).

- **WES / WGS:** High-yield first-line for undiagnosed neonatal hypotonia + developmental delay; trio analysis identifies de novo status.
- **Gene panels:** Intellectual disability / epilepsy / hypotonia NGS panels that include *PURA*.
- **Single-gene / targeted Sanger sequencing:** Feasible because the syndrome is **clinically recognizable** — *"The identification of one individual through targeted Sanger sequencing points towards the clinical recognisability of the syndrome"* (PMID: 29097605).
- **Chromosomal microarray (CMA):** Detects 5q31.3 (5q31.2q31.3) deletions encompassing *PURA* (PMID: 25439098; PMID: 26582469).

- Karyotyping/FISH: lower yield; CMA preferred for copy-number detection. Mitochondrial DNA and repeat-expansion testing: not applicable.

Variant interpretation. Reijnders et al. classified mutations using **3D in silico crystal-structure models** and performed computational facial-photograph analysis; genotype-phenotype analysis showed *"no significant correlation between mutation classes and disease severity"* (PMID: 29097605).

Clinical / ancillary tests. EEG (for seizures/seizure-like movements), brain MRI (often normal in point-mutation cases; more frequently abnormal in deletion carriers, PMID: 36376392), swallow/feeding assessment, sleep and respiratory evaluation, echocardiography and renal/urogenital imaging when malformations are suspected, and endocrine workup (suspected anterior pituitary dysregulation, PMID: 25342064).

Biomarkers. No specific circulating biomarker or metabolic signature; diagnosis is genetic. No methylation episignature established.

Differential diagnosis. Prader-Willi syndrome (neonatal hypotonia, feeding difficulty, hypersomnolence), Angelman syndrome, other genetic hypotonia/developmental-encephalopathy syndromes (e.g., *MECP2*, *FOXP1*, *TCF4*), congenital myopathies, and 5q31.3 microdeletion syndrome itself. The combination of neonatal hypotonia, hypersomnolence, excessive hiccups, and feeding difficulty is characteristic.

Screening. No newborn or carrier screening exists (de novo dominant disorder). Cascade testing of parents is generally uninformative for recurrence beyond germline-mosaicism counseling.

11. Outcome / Prognosis

- **Survival / life expectancy:** Many patients survive into adulthood; the oldest reported patient died at **age 26** from progressive respiratory failure (PMID: 26582469). No formal survival curves exist given rarity.
- **Mortality:** Main life-limiting risk is **respiratory complication**: *"her respiratory problems increased and eventually became severe enough to cause her death"* (PMID: 26582469). Neonatal apnea is a serious early risk.
- **Morbidity / disability:** Severe and lifelong. Most patients are **non-verbal**; many are **non-ambulatory**. Intellectual disability is moderate-to-severe.

- **Complications:** Seizures, aspiration/feeding failure (may require gastrostomy), recurrent apnea, orthopedic sequelae of hypotonia (scoliosis, hip issues), and organ-specific complications from congenital anomalies.
- **Recovery potential:** No recovery of core neurodevelopmental deficit; supportive care improves function and comfort. Some neonatal features (hiccups, apnea, hypothermia) attenuate with age.
- **Prognostic factors:** No validated molecular prognostic biomarker. Notably, **no genotype-phenotype correlation** guides prognosis (PMID: 29097605; PMID: 37204304). Respiratory status is the most important clinical prognostic determinant.

12. Treatment

There is no disease-modifying or curative therapy. Management is entirely **supportive, symptomatic, and multidisciplinary.**

Pharmacotherapy (symptomatic). - Anti-seizure medications for epilepsy (NCIT: Anticonvulsant Agent). Choice individualized; no PURA-specific regimen. - Management of movement disorders and sleep as needed. - No pharmacogenomic guidance specific to PURA syndrome.

Advanced therapeutics. No approved gene therapy, cell therapy, RNA-based therapy, or targeted therapy. Given the loss-of-function/haploinsufficiency mechanism and the essential dosage-sensitive role of Pur-alpha, **dosage-restoration strategies (gene supplementation, ASO-mediated upregulation)** are conceptually plausible future directions but remain preclinical/experimental (inferred).

Surgical / interventional. Gastrostomy for severe feeding difficulty (NCIT: Gastrostomy); corrective surgery for congenital heart or urogenital malformations; orthopedic intervention for scoliosis/hip.

Supportive and rehabilitative (core of care). - Feeding support: thickened feeds, NG/gastrostomy (NCIT: Nutritional Support). - Physical therapy for hypotonia/motor delay (NCIT: Physical Therapy). - Occupational therapy (NCIT: Occupational Therapy). - Speech therapy and **augmentative and alternative communication (AAC)** given near-universal absent speech (NCIT: Speech Therapy). - Respiratory monitoring/support (apnea management, ventilatory support as needed). - Sleep management for hypersomnolence.

Experimental treatments / clinical trials. No PURA-specific interventional trials with reported efficacy identified. Natural-history data collection and registries (PURA Syndrome Foundation) support future trial readiness.

Treatment strategy. Individualized, symptom-directed, delivered by a multidisciplinary team (neurology, genetics, gastroenterology, pulmonology, cardiology, PT/OT/speech, developmental pediatrics).

13. Prevention

- **Primary prevention:** Not possible — the disorder arises de novo. General reproductive-health measures do not prevent de novo mutation.
- **Secondary prevention:** Early molecular diagnosis enables early intervention (feeding support, apnea monitoring, developmental therapy, seizure management) that reduces morbidity.
- **Tertiary prevention:** Proactive management of complications — aspiration prevention, respiratory surveillance, seizure control, orthopedic monitoring.
- **Immunization / public health / environmental interventions:** Not applicable.
- **Genetic counseling:** Central. Explain de novo origin, **low (~1%) sibling recurrence risk** due to possible parental germline mosaicism, and reproductive options including **prenatal diagnosis and preimplantation genetic testing (PGT-M)** for a known familial variant.
- **Screening:** No population newborn or carrier screening (de novo dominant).

14. Other Species / Natural Disease

- **Orthologous gene:** Mouse *Pura* (NCBI Gene 19290); highly conserved across vertebrates. Pur-alpha is an evolutionarily conserved guanine-rich polynucleotide-binding protein; PC4-like PUR domains occur throughout all kingdoms of life ([PMID: 32476604](https://pubmed.ncbi.nlm.nih.gov/32476604/)).
- **Natural disease in other species:** No naturally occurring companion-animal or wildlife equivalent of PURA syndrome is documented (not identified in OMIA to date).
- **Comparative biology:** Disease mechanisms are studied via engineered models (below). Pur-alpha biology is deeply conserved; a baculovirus has acquired a functional PUR α -like

protein (VP1054) that binds GGN repeats, illustrating conservation of Pur nucleic-acid-binding function ([PMID: 23720732](#)).

- **Transmission / zoonotic potential:** Not applicable (genetic disorder).

15. Model Organisms

Mouse models are the principal system and recapitulate key features.

Model	Key phenotype	Human-relevant recapitulation	Reference
<i>Pura</i> heterozygous (+/–)	Decreased escape response to touch, limb/abdominal hypotonia, gait ataxia (wider steps, missteps), memory deficits (Barnes maze, novel object location), reduced neuron numbers in cerebellar vermis and hippocampal CA1–3	Models hypotonia, motor deficits, cognitive impairment, neuronal loss — the best dosage-matched model of human haploinsufficiency	PMID: 27651147
<i>Pura</i> null (–/–)	Born normal, then tremor at ~2 weeks, megalencephaly, axonal swellings, neurofilament hyperphosphorylation, prolonged postnatal proliferation of neuronal precursors (hippocampus, cerebellum), altered MAP2; some lines die postnatally	Models developmental/proliferation mechanism; more severe than human heterozygous state	PMID: 22010047

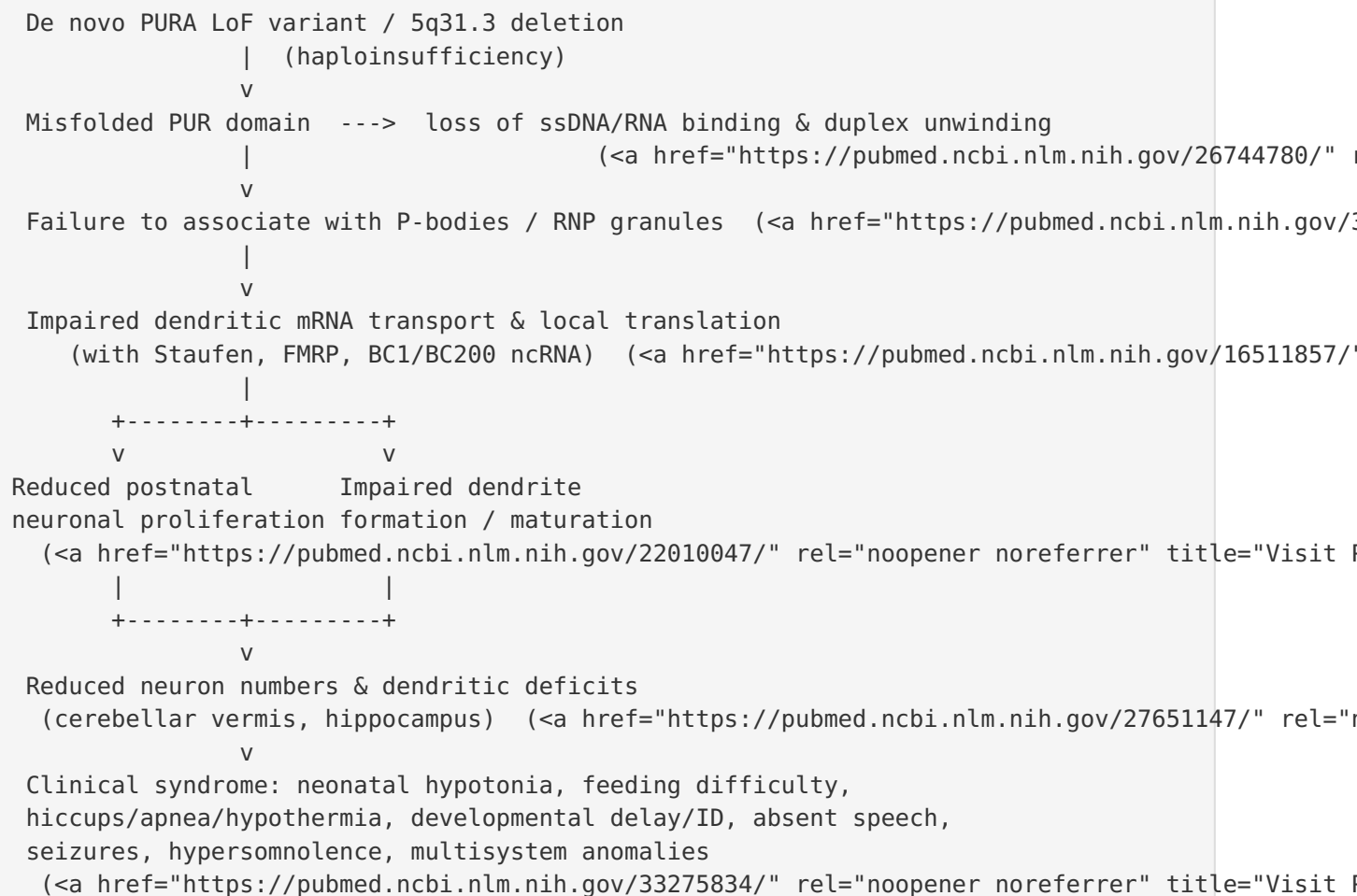
Supporting quotes: - "Standardized behavioral phenotyping revealed a decreased escape response to touch, limb and abdominal hypotonia, and gait abnormalities in heterozygous *Pur*-alpha (+/-) mice" ([PMID: 27651147](#)). - "revealed reduced numbers of neurons in general, as well as reduced numbers of *Pur*-alpha+-immunopositive neurons and dendrites in heterozygous *Pur*-alpha mice" ([PMID: 27651147](#)). - "lack of *Pura* prolongs the postnatal proliferation of neuronal precursor cells both in the hippocampus and in the cerebellum" ([PMID: 22010047](#)).

In vitro / cellular models. Patient-mutation cell systems demonstrated impaired PURA folding, disrupted RNA binding, and failed P-body localization ([PMID: 38655849](#)). Structural biology (crystallography of the DNA/RNA-binding domain with ssDNA) provides atomic-level models of mutation effects ([PMID: 26744780](#)).

Model limitations. The heterozygous mouse is milder than the human syndrome in some domains and does not fully capture absent speech, seizures, or the full multisystem spectrum. Null mice model developmental mechanisms but exceed the human haploinsufficient dosage state. iPSC-derived neurons/organoids are a promising, not-yet-exhaustively-characterized avenue (inferred).

Resources. MGI (*Pura*); Alliance of Genome Resources.

Mechanistic Model / Interpretation



Upstream vs downstream. The upstream lesion is the *PURA* variant and consequent protein misfolding/dosage loss; the proximal molecular defect is loss of nucleic-acid binding and RNP-granule/P-body association; downstream are the neurodevelopmental cellular deficits (proliferation, dendritogenesis) and, most distally, the clinical phenotype. The absence of a genotype-phenotype correlation suggests that once Pur-alpha function drops below a critical threshold, the phenotype is driven by haploinsufficiency generally rather than by the specific variant class.

Evidence Base

PMID	Title (abbrev.)	Role in this report
25439098	Mutations in <i>PURA</i> cause profound neonatal hypotonia... in 5q31.3 microdeletion syndrome	Establishes de novo <i>PURA</i> mutations as causal within the critical region
25342064	WES in family trios reveals de novo <i>PURA</i> mutations	Independent trio confirmation; core neonatal features; Pur repeat III hypothesis
26582469	Long-term follow-up, smallest 5q31.2q31.3 deletion	Confirms phenocritical role; documents adolescent deterioration and respiratory mortality
36376392	25 Mainland Chinese cohort	Phenotype frequencies; recurrent variants; MRI differences deletion vs point mutation
33275834	Expanding the phenotype (Phe233del)	Comprehensive multisystem phenotype; 78 individuals / 64 variants
37204304	New case, recurrent Phe233del	No reliable genotype-phenotype correlation; <i>Pura</i> functions
29097605	Clinical delineation in 32 individuals	Clinical recognizability; no genotype-phenotype correlation; 3D modeling
38655849	Mutations impair PUR-domain integrity & P-body association	Molecular mechanism (misfolding, RNA binding, P-bodies); ~650 patients
26744780	Structural basis of nucleic-acid recognition & unwinding	Crystal structure; molecular explanation of point mutations
32476604	PC4-like domain family	PUR-domain fold and nucleic-acid-binding properties
16511857	Role of Pur alpha in targeting mRNA in dendrites	Dendritic mRNA transport with Staufen/FMRP/BC1; essential for dendrite formation
22010047	Lack of Pur-alpha alters brain development	Null-mouse developmental mechanism, megalencephaly, proliferation
27651147		

PMID	Title (abbrev.)	Role in this report
	Memory deficits, gait ataxia, neuronal loss in Purα +/- mice	Heterozygous mouse recapitulation; neuronal loss
32035967	Pur-based peptide & C9orf72 G-quadruplex	Pur G-quadruplex biology; therapeutic-concept context
23720732	Baculovirus VP1054 as acquired PURα	Evolutionary conservation of Pur nucleic-acid-binding function

Limitations and Knowledge Gaps

1. **No formal epidemiology.** Prevalence/incidence are unknown; patient counts reflect ascertainment, not population frequency.
2. **No genotype-phenotype correlation.** Prognostic counseling cannot be refined by variant class ([PMID: 29097605](#); [PMID: 37204304](#)).
3. **Mechanism partly inferred from models.** The dendrite/proliferation defects and dosage sensitivity derive largely from mouse and in vitro systems; direct human neuropathology is limited.
4. **No biomarker.** No circulating, imaging, or epigenetic biomarker for diagnosis, severity, or progression.
5. **Limited long-term natural history.** Adult outcomes and the frequency of progressive respiratory decline are under-characterized (largely a single long-term case).
6. **No therapeutics.** No disease-modifying treatment or interventional trial data; dosage-restoration approaches are conceptual.
7. **Epigenetics / methylation signature** not established.
8. **iPSC/organoid human models** not yet exhaustively characterized.

Proposed Follow-up Experiments / Actions

1. **Prospective natural-history registry** (via PURA Syndrome Foundation) with standardized longitudinal capture of respiratory, seizure, feeding, communication, and mobility outcomes to define prognosis and trial endpoints.
 2. **iPSC-derived neurons and cortical/cerebellar organoids** carrying patient variants to quantify dendritogenesis, P-body dynamics, and translational dysregulation in a human context, validating the mouse-derived mechanism.
 3. **Transcriptome-wide identification of Pur-alpha-bound and dysregulated dendritic mRNAs** (CLIP-seq + ribosome profiling) in patient and model neurons to pinpoint downstream effectors.
 4. **Dosage-restoration proof-of-concept**: test AAV gene supplementation or ASO-mediated upregulation of *PURA* in heterozygous mice, given the haploinsufficiency mechanism and dosage sensitivity.
 5. **Structure-guided variant classification pipeline** using crystal-structure/AlphaFold models to standardize VUS interpretation and probe whether subtle folding-stability differences underlie the variable expressivity despite no categorical genotype-phenotype correlation.
 6. **Respiratory-focused clinical surveillance protocol** and apnea/sleep studies to reduce the principal source of mortality.
 7. **Search for a methylation epismutation** across a PURA cohort to enable epismutation-based diagnostic support.
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Report compiled from 8 confirmed findings across 15 reviewed primary papers. Evidence types: human clinical (cohorts, case series, trio exome studies), model organism (mouse knockout/heterozygous), in vitro (patient-mutation cell systems), and computational/structural (crystallography, 3D modeling).
